MEA OCPP 1.6 Compliance Test Report

Section 5: Reservation Check

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-11 12:07

Contents

1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct (no proxy)
Test PC	192.168.111.185 (alias 192.168.1.200/24 on enp3s0)
EV Plug Timeout	0 s per plug event
Expiry Wait	90 s (wait for CSMS to expire short reservation)
Test Date	2026-06-11 04:38:37 UTC
Results file	tex/vsecc_section5_results.json

Test Architecture

Reservation commands are issued via the MEA REST API.

- `POST /EV/cmd/chargepoint/reserve` with `chargepoint`, `connector`, `card_id`, and `duration`.
- `POST /EV/cmd/chargepoint/cancel` with `chargepoint` and `reservation_id`.

MQTT topics `vsecc/connector/+/status/#` and `vsecc/connector/+/ev/#` are observed for status changes.
Three reservation scenarios are tested:

1. **Flow 1 — Reserve → Cancel (5.1–5.5):** ReserveNow is sent for 5 minutes, then CancelReservation is issued. Charger should return to Available.
2. **Flow 2 — Reserve → Expiry (5.6–5.8):** ReserveNow is sent with a short duration. The CSMS expires the reservation automatically and the charger returns to Available.
3. **Flow 3 — Reserve → EV Plug → Charge (5.9–5.19):** ReserveNow is sent for 10 minutes, an EV is plugged in while the connector is reserved, RemoteStart initiates a charging session, and RemoteStop terminates it.

2 Section 5 Results

Summary: 6 PASS 0 FAIL 13 WARN 0 SKIP (19 total)

Item	Test Description	Detail / Evidence	Result
Flow 1: Reserve → Cancel — Items 5.1–5.5			
5.1	StatusNotification Available (initial)	`Available not observed on MQTT in 10 s` *Check CSMS event log; charger may already be available*	WARN
5.2	ReserveNow (5 min) → Accepted	`HTTP 200` *reservation_id not found in response*	PASS
5.3	StatusNotification Reserved (after ReserveNow)	`Reserved status not observed on MQTT in 15 s` *vSECC may use different status reporting for reservations*	WARN
5.4	CancelReservation (id=0) → Accepted	`HTTP 200`	PASS
5.5	StatusNotification Available (after cancel)	`Available not observed on MQTT in 15 s after cancel`	WARN
Flow 2: Reserve → Expiry — Items 5.6–5.8			
5.6	ReserveNow (short, for expiry test) → Accepted	`HTTP 200` *reservation_id not found*	PASS

Item	Test Description	Detail / Evidence	Result
5.7	Reservation expiry (wait 90s)	`Available not observed in 90s` *MEA CSMS may enforce a minimum reservation duration; expiry auto-cancelled*	WARN
5.8	StatusNotification Available (after expiry)	`Available not observed on MQTT in 15 s after expiry/cancel`	WARN
Flow 3: Reserve → EV Plug → RemoteStart → Charge → RemoteStop — Items 5.9–5.19			
5.9	ReserveNow (10 min, for charge session) → Accepted	`HTTP 200` *reservation_id not found*	PASS
5.10	StatusNotification Reserved (Flow 3)	`Reserved status not observed on MQTT in 15 s`	WARN
5.11	StatusNotification Reserved/Preparing (EV plug)	`EV_WAIT_SEC=0 | skipped` *Connect physical EV to connector and rerun*	WARN
5.12	RemoteStartTransaction → Accepted	`HTTP 200`	PASS
5.13	StartTransaction (triggered by RemoteStart)	`charging_session_state not observed on MQTT in 30 s`	WARN
5.14	StatusNotification Charging	`Charging status not observed on MQTT in 20 s`	WARN
5.15	MeterValues (during reservation charge)	`MeterValues not observed on MQTT in 15 s`	WARN
5.16	RemoteStopTransaction → Accepted	`HTTP 200`	PASS
5.17	StopTransaction (triggered by RemoteStop)	`session_state idle/finished not observed on MQTT in 25 s`	WARN
5.18	StatusNotification Finishing (EV still plugged)	`Finishing not observed on MQTT in 15 s` *Some implementations skip Finishing → go directly to Available*	WARN
5.19	StatusNotification Available (EV unplug)	`Available not observed on MQTT in 30 s | unplug EV`	WARN

3 Notes on Failures and Warnings

Reservation API

- The reserve endpoint is `POST /EV/cmd/chargepoint/reserve`. Payload keys: `"chargepoint"` (not `"chargepoint_id"`), `"connector"`, `"card_id"`, and `"duration"` (seconds).
- The cancel endpoint is `POST /EV/cmd/chargepoint/cancel`. Payload keys: `"chargepoint"` and `"reservation_id"`.
- The `reservation_id` is extracted from the ReserveNow response body at `data["result"]["reservation_id"]`.

StatusNotification observation

MQTT topic `vsecc/connector/1/status/...` reports `Reserved`, `Preparing`, `Charging`, `Finishing`, and `Available` states. Items that WARN on this topic indicate that the vSECC did not publish a state change within the observation window, or that the status key differs from the expected keyword.

Reservation expiry (Flow 2)

The MEA CSMS may enforce a minimum reservation duration longer than the requested 60 seconds. If the Available status is not observed within `EXPIRY_WAIT_SEC`, the test cancels the reservation automatically to unblock Flow 3. Rerun with a longer expiry wait or a shorter minimum enforced at the CSMS if this item WARNs.

EV-dependent items (Flow 3)

- **5.11** StatusNotification Reserved/Preparing: requires a physical EV to be connected to the reserved connector.
- **5.12–5.19** All depend on RemoteStart succeeding with an EV present and a valid reserved RFID tag.

Set `EV_WAIT_SEC > 0` in the test script to enable EV plug detection.

4 Dependency Map

Flow 1 5.1 → 5.2 (ReserveNow) → 5.3 → 5.4 (CancelReservation) → 5.5

Flow 2 5.6 (ReserveNow) → 5.7 (Expiry) → 5.8

Flow 3 5.9 (ReserveNow) → 5.10 → 5.11 (EV plug) → 5.12 (RemoteStart)
→ 5.13 → 5.14 → 5.15 (MeterValues) → 5.16 (RemoteStop)
→ 5.17 → 5.18 → 5.19 (Unplug)